

1. System Architecture

1.1 Architecture Overview

The system implements an AUTOSAR-based vehicle control architecture consisting of application Software Components, communication interfaces, ports, signals, sensors, and functional processing flows.

The application layer contains two primary Software Components:

- **EngineControl**
- **BrakeControl**

The Software Components communicate through AUTOSAR-defined ports and interfaces. Vehicle speed information is provided through the ISpeedData interface, while brake command information is exchanged through the IBrakeCommand interface.

The architecture separates application functionality from sensor and communication implementation. Software Components access required data through R-Ports and provide data through P-Ports.

1.2 Software Components

1.2.1 EngineControl

Software Component: EngineControl

EngineControl is responsible for processing vehicle speed information received from the speed sensing subsystem.

The component consumes vehicle speed data through the SpeedDataReceiver R-Port.

Responsibilities:

- **Receive vehicle speed information.**
- **Validate VehicleSpeed data.**
- **Process vehicle speed information.**
- **Detect invalid or unavailable speed data.**
- **Provide processed information to dependent functions.**

1.2.2 BrakeControl

Software Component: BrakeControl

BrakeControl is responsible for processing brake-related information and controlling the brake command based on vehicle operating conditions.

The component consumes required vehicle information through configured R-Ports and provides brake command information through the IBrakeCommand interface.

Responsibilities:

- Receive vehicle speed information.
 - Receive brake-related information from the brake sensing subsystem.
 - Process brake control conditions.
 - Generate BrakePressure information.
 - Detect invalid brake sensor data.
 - Provide brake command information to the communication layer.
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1.3 AUTOSAR Interfaces

1.3.1 ISpeedData

Interface: ISpeedData

Communication Type: Sender/Receiver

ISpeedData is an AUTOSAR Sender/Receiver interface used to transfer vehicle speed information between the speed data provider and consuming Software Components.

The interface contains the VehicleSpeed data element.

Data Element

Interface	Data Element	Data Type	Unit	Description
ISpeedData	VehicleSpeed	uint16	km/h	Current vehicle speed

Interface Usage

The ISpeedData interface is used by:

- SpeedDataProvider
- SpeedDataReceiver

The provider publishes VehicleSpeed information, while the receiver consumes the information.

1.3.2 IBrakeCommand

Interface: IBrakeCommand

Communication Type: Sender/Receiver

IBrakeCommand is an AUTOSAR interface used to transfer brake command information from the brake control application to the dependent communication or actuator subsystem.

The interface contains the **BrakePressure** data element.

Data Element

Interface	Data Element	Data Type	Unit	Description
IBrakeCommand	BrakePressure	uint16	bar	Requested brake pressure

Interface Usage

The **IBrakeCommand** interface is associated with the brake control processing function.

BrakeControl generates the **BrakePressure** value based on the current vehicle operating conditions.

1.4 Port Architecture

1.4.1 SpeedDataProvider

Port Type: P-Port

P-Port: SpeedDataProvider

Interface: ISpeedData

The **SpeedDataProvider** P-Port provides vehicle speed information to other Software Components through the **ISpeedData** interface.

The port publishes the **VehicleSpeed** data element.

Port Definition

Attribute	Value
Port Name	SpeedDataProvider
Port Type	P-Port
Interface	ISpeedData
Data Element	VehicleSpeed
Direction	Provided

Attribute	Value
Source	Speed sensing subsystem

1.4.2 SpeedDataReceiver

Port Type: R-Port

R-Port: SpeedDataReceiver

Interface: ISpeedData

The SpeedDataReceiver R-Port consumes vehicle speed information from the ISpeedData interface.

The port provides VehicleSpeed information to application functionality that requires vehicle speed.

Port Definition

Attribute	Value
Port Name	SpeedDataReceiver
Port Type	R-Port
Interface	ISpeedData
Data Element	VehicleSpeed
Direction	Required
Consumer	EngineControl / BrakeControl

1.5 Signal Architecture

1.5.1 VehicleSpeed

Signal: VehicleSpeed

Interface: ISpeedData

Data Type: uint16

Unit: km/h

VehicleSpeed represents the current speed of the vehicle.

The value is acquired from the speed sensing subsystem and published through the SpeedDataProvider P-Port.

The SpeedDataReceiver R-Port consumes the signal for application processing.

VehicleSpeed Properties

Property	Value
Signal Name	VehicleSpeed
Interface	ISpeedData
Data Type	uint16
Unit	km/h
Minimum Value	0
Maximum Value	300
Communication Type Periodic	
Cycle Time	10 ms
Maximum Latency	20 ms

1.5.2 BrakePressure

Signal: BrakePressure

Interface: IBrakeCommand

Data Type: uint16

Unit: bar

BrakePressure represents the requested brake pressure generated by the BrakeControl Software Component.

The signal is generated after evaluating vehicle speed and brake-control conditions.

BrakePressure Properties

Property	Value
Signal Name	BrakePressure

Property	Value
Interface	IBrakeCommand
Data Type	uint16
Unit	bar
Minimum Value	0
Maximum Value	200
Communication Type	Periodic / Event-driven
Validity	Validated before transmission

1.6 Sensor Dependencies

1.6.1 EngineControl Dependency

Software Component: EngineControl

Dependency: SpeedSensor

EngineControl depends on SpeedSensor for obtaining the current vehicle speed.

The dependency is represented as:

SpeedSensor → VehicleSpeed → ISpeedData → SpeedDataReceiver → EngineControl

EngineControl shall not directly access the physical SpeedSensor hardware. Sensor information shall be made available through the appropriate AUTOSAR abstraction and communication mechanisms.

Dependency Definition

Consumer	Dependency Data	Purpose
EngineControl	SpeedSensor VehicleSpeed	Vehicle speed processing

1.6.2 BrakeControl Dependency

Software Component: BrakeControl

Dependency: BrakeSensor

BrakeControl depends on BrakeSensor for obtaining brake-related information required during brake control processing.

The dependency is represented as:

BrakeSensor → Brake Information → BrakeControl

BrakeControl shall process validated sensor information before generating the brake command.

Dependency Definition

Consumer	Dependency Data	Purpose
BrakeControl	BrakeSensor Brake status / pressure information	Brake control processing

1.7 Software Component Relationships

The major architectural relationships are summarized below.

Source	Relationship	Target	Data / Interface
SpeedSensor	Provides	SpeedDataProvider	VehicleSpeed
SpeedDataProvider	Publishes through	ISpeedData	VehicleSpeed
ISpeedData	Supplies	SpeedDataReceiver	VehicleSpeed
SpeedDataReceiver	Provides data to	EngineControl	VehicleSpeed
SpeedDataReceiver	Provides data to	BrakeControl	VehicleSpeed
BrakeSensor	Provides data to	BrakeControl	Brake information
BrakeControl	Generates	IBrakeCommand	BrakePressure
IBrakeCommand	Carries	Brake command subsystem	BrakePressure

1.8 Functional Flow

1.8.1 Functional Flow: Vehicle Speed Processing

The Vehicle Speed Processing function obtains vehicle speed from the SpeedSensor and makes the information available to EngineControl through the AUTOSAR communication architecture.

Processing Sequence

1. SpeedSensor measures the current vehicle speed.

2. The speed information is converted into the VehicleSpeed signal.
3. VehicleSpeed is published through the SpeedDataProvider P-Port.
4. SpeedDataProvider uses the ISpeedData interface.
5. The SpeedDataReceiver R-Port receives VehicleSpeed.
6. EngineControl consumes the VehicleSpeed information.
7. EngineControl validates the received value.
8. EngineControl performs vehicle speed processing.
9. Invalid or unavailable data is handled according to the configured error-handling strategy.

Functional Flow

SpeedSensor → VehicleSpeed → SpeedDataProvider → ISpeedData → SpeedDataReceiver → EngineControl

1.8.2 Functional Flow: Brake Control Processing

The Brake Control Processing function uses vehicle and brake sensor information to generate an appropriate brake command.

Processing Sequence

1. BrakeSensor provides brake-related information.
2. SpeedSensor provides the current vehicle speed.
3. VehicleSpeed is transferred through ISpeedData.
4. SpeedDataReceiver receives VehicleSpeed.
5. BrakeControl consumes the required input information.
6. BrakeControl validates the received data.
7. BrakeControl evaluates brake-control conditions.
8. BrakeControl calculates the required BrakePressure.
9. BrakePressure is published through the IBrakeCommand interface.
10. The brake command is transferred to the dependent brake actuator or control subsystem.

Functional Flow

SpeedSensor → VehicleSpeed → ISpeedData → SpeedDataReceiver → BrakeControl

BrakeSensor → BrakeControl → BrakePressure → IBrakeCommand → Brake Actuator

1.9 End-to-End Architecture

The overall application architecture can be represented as:

SpeedSensor → SpeedDataProvider → ISpeedData → SpeedDataReceiver → EngineControl

SpeedSensor → SpeedDataProvider → ISpeedData → SpeedDataReceiver → BrakeControl

BrakeSensor → BrakeControl → BrakePressure → IBrakeCommand → Brake Actuator

The architecture establishes a clear separation between sensors, communication interfaces, AUTOSAR ports, application Software Components, and actuator control.

1.10 Architecture Constraints

The following constraints shall be maintained:

- **EngineControl shall obtain vehicle speed through the defined AUTOSAR communication interface.**
- **BrakeControl shall use validated sensor information.**
- **VehicleSpeed shall be associated with ISpeedData.**
- **BrakePressure shall be associated with IBrakeCommand.**
- **SpeedDataProvider shall be configured as a P-Port.**
- **SpeedDataReceiver shall be configured as an R-Port.**
- **Application Software Components shall not directly access sensor hardware.**
- **Application Software Components shall not directly access hardware drivers.**
- **Interface definitions shall remain consistent with port definitions.**
- **Signal data types shall remain consistent between provider and consumer.**
- **Changes to interfaces or ports shall undergo impact analysis.**